Cape Peninsula University of Technology

SOFTWARE DESIGN 2

GROUP 32 AUTOVAULT TESTING EVIDENCE

Group members

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AUTOVAULT: TESTING EVIDENCE

1. Purpose

This report demonstrates our testing procedure of AutoVault including test cases, input values we applied (both typical and boundary ones), our expectations, discovered bugs and their elimination as well as output data sample. The design is described in the Software Design Report whereas usage guidelines are provided in the User Guide.

2. How we tested

2.1 Automatic test runner

AutoVaultTests runs 25 test cases against the actual BackupManager without writing any code. AutoVaultTests creates its own directories (test-data), runs all test cases, and shows PASS/FAIL along with observed output data. Results are also written into test-results.txt. This tester makes sure that the actual jobs.dat file remains untouched, since AutoVaultTests moves it elsewhere before running test cases and restores it afterwards. There is an additional test case (A1) for automatic backups which takes around 70 seconds to complete, thus it is not included by default.

2.2 Manual tests

Some things cannot be checked by the runner, such as what the user sees in the console menu and in the window. We tested these by hand (Section 5).

2.3 Logging tests

The logging class was independently tested in the QA window (AutoVaultRole6Prototype, “Run QA Tests” button). They are listed in Section 6. The runner also verifies that actual backups and actual error messages write lines in AutoVault.log.

2.4 Test environment

Item	**Value**
Java	Java 21
IDE	Apache NetBeans 25
Operating system	Windows 11
Date of the test run	5 November 2026
Run by:	Aphiwe Mtshali

3. Normal input test cases

ID	Test	Input	Expected result	Result
N1	Add a valid job	Name TestJob, real source and backup folders, 1 minute	The job appears in the job list	PASS
N2	First backup	Back up the new job	One version is created	PASS
N3	Backup after an edit	Edit report.txt, back up again	A second version is created	PASS
N4	Version order	List the versions	Versions are listed oldest first	PASS
N5	Restore the	Restore version 1 into a	report.txt has the first-draft	PASS

	oldest version	new folder	text	
N6	Retention (jobs)	8 backups, each after an edit	The job keeps only the newest 5 versions	PASS
N7	Retention (disk)	Count the ver folders in the backup folder	Deleted versions' folders are gone: 5 folders for 5 versions	PASS
N8	Save and load	Save, start a new manager, load	The job and its versions are still there	PASS
N9	Backup is logged	Edit a file, back up, compare the size of AutoVault.log	The log file is bigger after the backup	PASS
N10	Special file names	Folder with spaces, file "my report (final).txt"	Backed up and restored correctly	PASS

4. Edge case test cases

ID	Test	Input	Expected result	Result
E1	Empty job name	Name made of three spaces	Rejected: "Job name must not be empty."	PASS
E2	Missing folder	Source folder = null	Rejected: source and destination folders are required	PASS
E3	Interval 0	Interval = 0	Rejected: "Interval must be at least 1 minute."	PASS
E4	Negative interval	Interval = -5	Rejected: "Interval must be at least 1 minute."	PASS
E5	Same folder	Source = destination	Rejected: must be different folders	PASS
E6	Destination inside source	Destination = source\ backups	Rejected: cannot be inside the source folder	PASS
E7	Backup of an unknown job	A job id that does not exist	Returns null, "Job not found", no crash	PASS
E8	Versions of an unknown job	A job id that does not exist	Returns null, no crash	PASS
E9	Restore an unknown version	A version id that does not exist	"Version not found", no crash	PASS
E10	Source folder deleted	Delete the folder, then back up	The backup fails safely, no crash	PASS
E11	Empty source folder	Back up a folder with no files	No crash	PASS
E12	No changes between backups	Two backups in a row, no edit	The second is skipped: "No changes ... Nothing to do.", still 1 version	PASS
E13	Restore to a new folder	Target folder does not exist yet	The folder is created and the file restored	PASS
E14	Removed job	Remove the job, then back up	Returns null, no crash	PASS
E15	Failed backup is logged	Repeat E10 and compare the size of	The log file is bigger after the failed backup	PASS

		AutoVault.log		

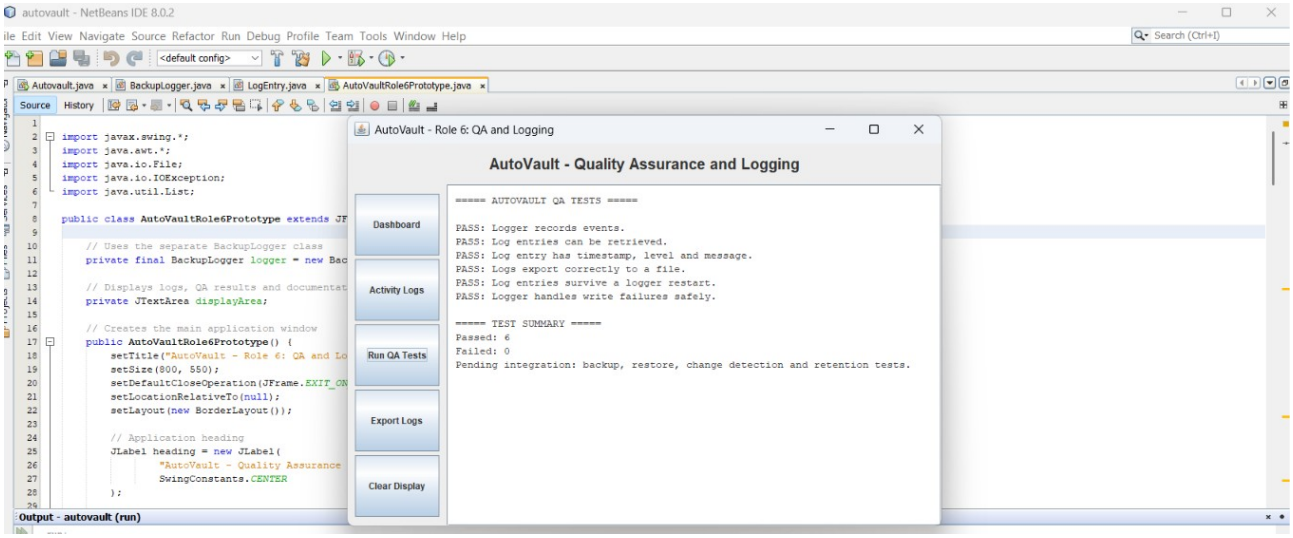
5. Manual tests

5.1 Window

ID	What we do	Expected result	Result
W1	Open the window	The table shows the saved jobs and the status line that says “Ready.”	The table shows the saved jobs
W2	Click Backup now with no job selected	A message asks us to select a job in the table first	Please select a job in a table first
W3	Add job with a source folder that does not exist	An error box says the source folder does not exist	Error box says the source folder does not exist
W4	Add job with an empty name	An error box says the job name must not be empty	Error box says the job name must not be empty
W5	Versions / Restore on a job with no versions	A message says the job has no versions yet	This job has no versions yet
W6	Backup now twice with no change in between	The status line says there are no changes and nothing to do	No changes since last backup, nothing to do.
W7	Remove job, then choose No, then try again and choose Yes	No keeps the job; Yes removes it	No keeps the job; Yes removes it
W8	Add a job in the console menu while the window is open	The job appears in the window table within 5 seconds	The job appears in the window table within 5 seconds
W9	Close the window	The jobs are saved and the program ends	The jobs are saved and the program ends

6. Logging tests (QA window)

These tests are run with the “Run QA Tests” button in AutoVaultRole6Prototype. Each one prints PASS or FAIL in the window.



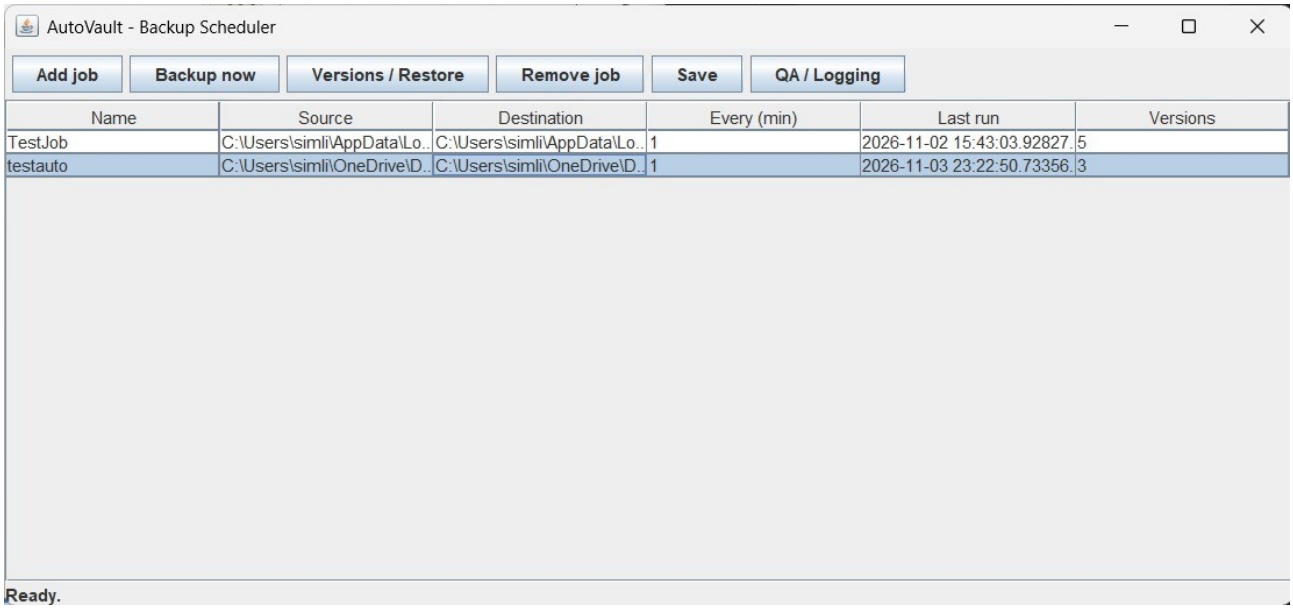
ID	What it checks	Input	Result
L1	The logger records a new event	One test entry is logged	PASS
L2	Log entries can be read back	Read all entries	PASS
L3	The newest entry has a time, a level and a message	Read the last entry	PASS
L4	The log can be exported to a file	Export, then read the file	PASS
L5	Entries survive a restart of the logger	Write, create a new logger, read	PASS
L6	A write failure is handled safely	Use a folder as the log file	PASS

7. Defects found and fixed

Testing found four defects. All four were fixed and then tested again.

Defect	What went wrong	Tested again by	Result
Old versions stayed in the job	The retention policy deleted old version folders from the disk, but the job still listed those versions.	N6 and N7	PASS
Crash when a backup failed	A failed backup returned nothing and the program did not handle it.	E10 and E15	PASS
Endless loop when the window opened	show() called setVisible, which called show() again.	W1	PASS
Wrong message when nothing had changed	The window showed an incorrect status when there was nothing to back up.	E12 and W6	PASS

8. Sample output




```
run:
=== AutoVault Test Run ===

--- Normal input ---
PASS  N1  Add a valid job
        observed: jobs in manager: 1
Backed up: notes.txt
Backed up: report.txt
Backup completed.
PASS  N2  First backup creates a version
        observed: versions: 1
Backed up: notes.txt
Backed up: report.txt
Backup completed.
PASS  N3  Backup after editing a file creates a second version
        observed: versions: 2
PASS  N4  Versions are listed oldest to newest
        observed: first=20261106_003954_247, last=20261106_003955_382
Restored folder:
Restored file: notes.txt
Restored file: report.txt
Restore completed successfully.
Restore finished.
PASS  N5  Restoring the oldest version brings back the old text
        observed: restored report.txt says: Report - first draft
Backed up: notes.txt
Backed up: report.txt
Backup completed.
Backed up: notes.txt
Backed up: report.txt
Backup completed.
```

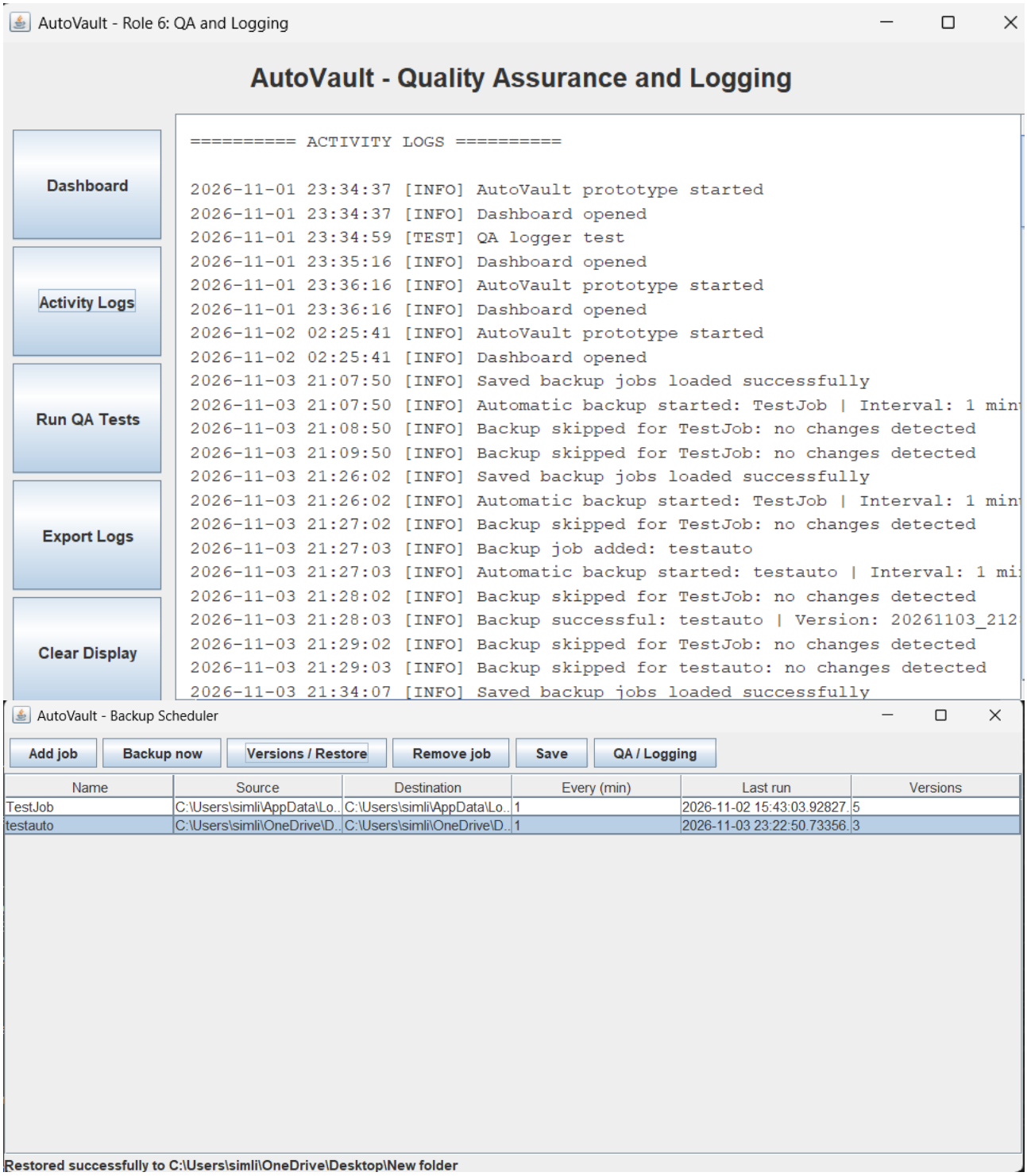
```
Backed up: notes.txt
Backed up: report.txt
Backup completed.
Backed up: notes.txt
Backed up: report.txt
Backup completed.
[Retention] Removed old version: 20261106_003954_247 | 2026-11-06 00:39:54 | 2 file(s) | 33 B
Backed up: notes.txt
Backed up: report.txt
Backup completed.
[Retention] Removed old version: 20261106_003955_382 | 2026-11-06 00:39:55 | 2 file(s) | 43 B
Backed up: notes.txt
Backed up: report.txt
Backup completed.
[Retention] Removed old version: 20261106_003956_529 | 2026-11-06 00:39:56 | 2 file(s) | 29 B
PASS  N6  Retention keeps only the newest 5 versions after 8 backups
        observed: versions kept: 5
PASS  N7  Deleted versions' folders are removed from the disk
        observed: version folders on disk: 5, versions in the job: 5
PASS  N8  Jobs and versions survive a restart (save and load)
        observed: after reload: 1 job(s), 5 version(s)
Backed up: notes.txt
Backed up: report.txt
Backup completed.
[Retention] Removed old version: 20261106_003957_657 | 2026-11-06 00:39:57 | 2 file(s) | 29 B
PASS  N9  A backup is recorded in AutoVault.log
        observed: AutoVault.log size: 16753 bytes before, 16914 bytes after
Backed up: my report (final).txt
Backup completed.
Restored folder:
Restored file: my report (final).txt
```

```
Restore completed successfully.
Restore finished.
PASS  N10  Names with spaces and brackets are backed up and restored
        observed: restored file found: true

--- Edge input ---
PASS  E1  Empty job name is rejected
        observed: message: "Job name must not be empty."
PASS  E2  Missing folder (null) is rejected
        observed: message: "Source and destination folders are required."
PASS  E3  Interval of 0 minutes is rejected
        observed: message: "Interval must be at least 1 minute."
PASS  E4  Negative interval is rejected
        observed: message: "Interval must be at least 1 minute."
PASS  E5  Source and destination the same folder is rejected
        observed: message: "Source and destination must be different folders."
PASS  E6  Destination inside the source folder is rejected
        observed: message: "Destination cannot be inside the source folder."
Job not found: no-such-job
PASS  E7  Backup of an unknown job returns null without crashing
        observed: result: null
Job not found: no-such-job
PASS  E8  Listing versions of an unknown job returns null without crashing
        observed: result: null
Backed up: notes.txt
Backed up: report.txt
Backup completed.
Version not found: no-such-version
PASS  E9  Restoring an unknown version does not crash
        observed: the program printed 'Version not found' and carried on
Source folder does not exist: C:\Users\simli\AppData\Local\Temp\autovault-tests-1793918393939\vanishing\source
```

```
Source folder does not exist: C:\Users\simli\AppData\Local\Temp\autovault-tests-1793918393939\vanishing\source
PASS  E10  Backup when the source folder was deleted fails safely
        observed: result: null
PASS  E15  That failed backup is recorded in AutoVault.log
        observed: AutoVault.log size: 17664 bytes before, 17839 bytes after
Backup completed.
PASS  E11  Backup of an empty folder does not crash
        observed: result: version created
Backed up: notes.txt
Backed up: report.txt
Backup completed.
No changes since the last backup.
PASS  E12  A second backup with no changes is skipped
        observed: 1 version(s) after two backups with no edit in between
Restored folder:
Restored file: notes.txt
Restored file: report.txt
Restore completed successfully.
Restore finished.
PASS  E13  Restoring into a folder that does not exist yet works
        observed: restored file found: true
Job not found: 66b6311d-55ae-4109-a822-714b316cc8c1
PASS  E14  A removed job cannot be backed up
        observed: jobs left: 3

=== SUMMARY ===
Passed: 25
Failed: 0
Results saved to C:\Users\simli\OneDrive\Documentos\NetBeansProjects\AutoV\test-results.txt
BUILD SUCCESSFUL (total time: 11 seconds)
```



9. Results summary

Group	Tests	Passed	Failed	Notes
Normal input (N1 to N10)	10	10	0	First run: 9 passed, 1 failed (N9). Latest run: all passed..

Edge cases (E1 to E15)	15	14	1	First run: 14 passed, 1 failed (E15). Latest run: all passed
Window (W1 to W9)	9	9	0	All tests passed
Logging (L1 to L6)	6	6	0	All tests passed on the first try
Total	40	40	0	All the tests passed

What we did not test

- Automatic backups over a long time. Test A1 only checks one scheduled run.
- Very large folders and very long file paths.
- Two threads using the job list at exactly the same moment. This is a known limit of BackupManager.